ECA56 Advanced Wireless Communication Systems

Assignment – 2

CO2: Analyze LTE and 5G wireless network architectures; including carrier aggregation, MIMO techniques, beamforming strategies, and service-based core network functional entities.

CO2 Engineering System Architecture Evaluation Assignment

5G NR architecture, cellular network planning, spectrum allocation, radio resource management and simulation-based performance validation

Assessment format	Duration	Maximum marks	Bloom levels
Open-ended engineering design report	1 week	100	K4 Analyze · K5 Evaluate · K6 Create

Engineering design problem

A mobile network operator plans to design and optimize a 5G New Radio (NR) network for a densely populated urban area to improve throughput, spectral efficiency and received signal quality. The existing baseline network uses a single 20 MHz carrier, 2×2 MIMO and no adaptive beamforming. Develop an engineering report that evaluates 5G NR architecture, carrier aggregation, higher-order MIMO, beamforming techniques, spectrum utilization, interference management and network performance optimization. Use MATLAB, Python or another suitable simulation environment to compare the baseline and proposed network configurations under common assumptions. Apply quantitative analysis to justify the final design while considering traffic demand, spectrum limitations, implementation complexity and realistic performance losses. Clearly distinguish theoretical calculations from simulation-based or measured results.

Design area	Supplied case data
Baseline 5G NR Network	Existing carrier bandwidth = 20 MHz; MIMO configuration = 2×2; adaptive beamforming is not used. Evaluate baseline throughput, spectral efficiency and received signal quality.
Carrier aggregation	Three component carriers of 20 MHz each; usable bandwidth efficiency = 90% per carrier. Evaluate aggregate bandwidth and throughput improvement.
MIMO configuration	Proposed configuration = 4×4 MIMO; SU-MIMO supports 2 spatial layers; MUMIMO schedules 4 users with 1 spatial layer per user. Compare spatial multiplexing and user capacity.

Beamforming	Eight antenna elements; inter-element spacing = \(\lambda/2\). Compare analog, digital and hybrid beamforming in terms of gain, interference reduction, complexity and hardware requirements.
5G NR air interface	Available bandwidth = 100 MHz; spectral efficiency = 3 bit/s/Hz; target aggregate throughput = 250 Mbps; subcarrier spacing = 30 kHz.
Simulation and Performance Validation	Compare baseline and proposed configurations using MATLAB, Python or another suitable simulation environment. Evaluate throughput, spectral efficiency, received signal quality, interference and resource utilization under common assumptions.

Stage 1 – Problem Formulation and Baseline Network Analysis (15 marks)

Bloom's level: BL4 – Analyze

Given the following network parameters, establish a baseline configuration and analyze its limitations.

Parameter	Given value
Operating frequency	3.5 GHz
Available bandwidth	100 MHz maximum
Baseline carrier bandwidth	20 MHz
Baseline MIMO configuration	2×2
Number of active users	100
Average user throughput requirement	10 Mbps
Target cell-edge received power	At least −90 dBm
Assumed baseline spectral efficiency	2 bits/s/Hz

Questions

1. Calculate the aggregate user throughput demand and the theoretical baseline throughput. Determine whether the baseline configuration satisfies the demand. (4 marks)

2. Identify at least four engineering constraints affecting the design, including spectrum availability, interference, user density and implementation complexity. (4 marks)

3. Establish a baseline simulation model, state all assumptions and identify the performance metrics to be measured. (4 marks)

4. Formulate two measurable design objectives and two technical constraints for the proposed network. (3 marks)

Stage 2 – Carrier Aggregation and MIMO Design (20 marks)

Bloom's levels: BL4–BL5

The network operator can configure up to 100 MHz of aggregated spectrum. Consider the following candidate configurations:

- Configuration A: 20 MHz, 2×2 MIMO.

- Configuration B: 40 MHz, 4×4 MIMO.

- Configuration C: 80 MHz, 4×4 MIMO.

- Configuration D: 100 MHz, 8×8 MIMO.

For preliminary calculations, assume spectral efficiency of 2 bits/s/Hz for the baseline and 4 bits/s/Hz for the proposed configurations. Apply a 20% implementation and protocol overhead to all configurations.

Questions

1. Develop the mathematical model for aggregate throughput with carrier aggregation and spatial multiplexing. (4 marks)

2. Calculate effective throughputs for all four configurations using the stated assumptions. (6 marks)

3. Evaluate whether each configuration can meet the aggregate demand of 1 Gbps for 100 users, each requiring an average of 10 Mbps. (4 marks)

4. Select a configuration that satisfies the bandwidth constraint and explain the trade-off between throughput, spectrum consumption and MIMO complexity. (6 marks)

Stage 3 – Beamforming Design and Link-Budget Validation (20 marks)

Bloom's levels: BL4–BL5

Design a beamforming subsystem operating at 3.5 GHz. Compare a 4-element array, a 16-element array and a 64-element array using a uniform linear array (ULA) with half-wavelength element spacing.

For initial analytical evaluation, use the ideal array-gain approximation $G=10\log10(N)$, where N is the number of elements. Assume the single-element received power is −100 dBm and the target received power is at least −90 dBm.

Questions

1. Calculate the wavelength and half-wavelength antenna spacing at 3.5 GHz. (4 marks)
2. Calculate the ideal array gain and predicted received power for all three array sizes. (6 marks)
3. Evaluate which arrays satisfy the target received-power requirement under ideal assumptions. (3 marks)
4. Design a simulation to compare beam patterns and beam steering at different angles. Include the effect of at least one non-ideal factor, such as phase errors or element losses. (4 marks)
5. Justify the selected array by considering received power, beamwidth, implementation complexity and practical limitations. (3 marks)

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Stage 5 – Simulation Validation, Performance Comparison and Engineering Decision (20 marks) Bloom's levels: BL5–BL6

Validate your proposed network against the baseline using the same traffic demand, channel assumptions and simulation conditions. Your results must demonstrate whether the proposed design achieves measurable improvement rather than relying only on theoretical calculations.

Questions

1. Implement the baseline and proposed configurations in MATLAB, Python or an equivalent simulation environment. Explain the simulation procedure, input parameters and assumptions so that the experiment can be reproduced. (4 marks)

2. Generate and interpret at least four performance plots: effective throughput versus bandwidth, throughput versus number of MIMO streams, received power versus number of antenna elements, and performance under different channel or interference conditions. (5 marks)

3. Compare the baseline and proposed designs using at least four metrics: effective throughput, spectral efficiency, received power, and performance relative to the user-demand requirement. Present the results in a comparison table. (4 marks)

4. Conduct a sensitivity analysis by changing at least two design parameters, such as user load, overhead, channel quality or antenna phase error. Identify when the proposed design no longer meets its target. (4 marks)

5. Recommend the final network configuration and justify your decision using simulation evidence, engineering trade-offs, limitations and possible future improvements. (3 marks) Submission requirements Students must submit:

1. Problem formulation, assumptions and engineering constraints.

2. Baseline and proposed 5G NR network architecture diagrams.

3. Mathematical calculations and design justification.

4. Complete, executable simulation code and input parameters.

5. At least four performance plots and a baseline-versus-proposed comparison table.

6. Sensitivity analysis, limitations, conclusions and references.

Submission requirements

Submit one engineering design report organized as: requirements → architecture analysis → alternatives → calculations → design selection → verification → recommendation. Include labeled architecture and radio diagrams, equations with units, comparison tables, stated assumptions, and a brief interpretation beside every major result. Any software or external technical sources used must be identified. Unsupported diagrams or software-generated numbers without engineering reasoning will not receive full credit.Assessment rubric

Assessment criterion	Marks	Primary Bloom level
Problem formulation and baseline analysis	15	K4 / K5
Carrier aggregation and SU-MIMO/MU-MIMO analysis	15	K4 / K5
Beamforming, beam steering and antenna-spacing calculations	15	K4 / K5
5G NR air interface, gNodeB architecture and RAN functional split	15	K5 / K6
RRC operation, states and connection establishment	15	K4 / K5
Simulation validation and engineering decision	25	K5 / K6
TOTAL	100	

Performance expectations

Excellent	Good	Satisfactory	Needs improvement
The student systematically analyses the 5G NR network requirements, compares carrier aggregation, higher-order MIMO and beamforming alternatives, develops an optimized network design, validates performance through simulation and quantitative metrics, and critically evaluates engineering constraints and design trade-offs.	The student demonstrates a correct 5G NR design methodology, implements appropriate carrier aggregation, MIMO and beamforming techniques, and validates most performance requirements, with minor errors or limited justification of design choices.	The student develops a workable 5G NR network model and obtains basic simulation results but demonstrates limited comparison of alternative configurations, incomplete performance validation, or insufficient justification of engineering decisions.	The student primarily reproduces theoretical calculations or simulation outputs without demonstrating effective network design, appropriate selection of carrier aggregation, MIMO and beamforming techniques, quantitative performance validation, or critical evaluation of engineering constraints.

ANSWER

Stage 1 – Problem Formulation and Baseline Network Analysis

1. Calculate aggregate user throughput demand and theoretical baseline throughput.

Given:

- Number of active users = 100

- Average user throughput = 10 Mbps
- Baseline bandwidth = 20 MHz
- Baseline spectral efficiency = 2 bits/s/Hz
- MIMO configuration = 2×2

Step 1: Aggregate user throughput demand

Formula:

$$T_{\text{demand}}=N\times T_{\text{user}}$$

$$=100\times10$$

$$\boxed{T_{\text{demand}}=1000\ \text{Mbps}=1\ \text{Gbps}}$$

Step 2: Theoretical baseline throughput

Formula:

$$T=B\times SE$$

Where:

- B = Bandwidth
- SE = Spectral efficiency

Substitute the values:

$$T=20\times2$$

$$\boxed{T=40\ \text{Mbps}}$$

Step 3: Comparison

$$40\ \text{Mbps} < 1000\ \text{Mbps}$$

Therefore, the baseline network cannot satisfy the total user demand.

Result: The baseline network provides only 40 Mbps, while the required throughput is 1000 Mbps. Hence, network improvement is necessary.

2. Identify at least four engineering constraints affecting the design.

1. Spectrum Availability: Limited bandwidth of 100 MHz restricts the maximum achievable throughput.
2. Interference: Signals from nearby cells may interfere with each other and reduce network performance.
3. User Density: A large number of users sharing the same network resources can reduce individual throughput.
4. Implementation Complexity: Higher-order MIMO and beamforming require additional antennas, hardware and processing power.
5. Signal Quality: Obstacles and distance from the base station can reduce received signal strength.

3. Establish a baseline simulation model, state assumptions and identify performance metrics.

Baseline simulation model:

The baseline network is designed using a 20 MHz carrier bandwidth and 2×2 MIMO configuration.

Assumptions:

- Operating frequency = 3.5 GHz
- Bandwidth = 20 MHz
- Number of users = 100
- Spectral efficiency = 2 bits/s/Hz
- MIMO = 2×2
- Average user demand = 10 Mbps

Performance metrics:

1. Throughput – Total data transmitted per second.

2. Spectral efficiency – Amount of data transmitted per unit bandwidth.
3. Received power – Strength of the received signal.
4. Interference – Unwanted signals affecting communication.
5. Resource utilization – Percentage of available network resources used.

4. Formulate two measurable design objectives and two technical constraints.

Design objectives:

1. Achieve an aggregate network throughput of at least 1 Gbps.
2. Maintain cell-edge received power of at least −90 dBm.

Technical constraints:

1. Available spectrum must not exceed 100 MHz.
2. The network must operate within practical hardware and implementation limitations.

Stage 2 – Carrier Aggregation and MIMO Design (20 Marks)

1. Develop the mathematical model for aggregate throughput with carrier aggregation and spatial multiplexing.

Carrier Aggregation (CA) combines multiple component carriers to increase the available bandwidth.
MIMO uses multiple antennas to transmit multiple data streams simultaneously.
Throughput formula:
$$ T_{\text{effective}}= B\times SE\times L\times(1-O) $$
Where:

- B = Aggregate bandwidth
- SE = Spectral efficiency per spatial layer
- L = Number of simultaneous spatial layers
- O = Implementation and protocol overhead

Given:
$$ O=20\%=0.2 $$
Therefore:
$$ T_{\text{effective}}=B\times SE\times L\times0.8 $$
For the preliminary calculations, spatial efficiency is treated as per-layer efficiency to account separately for spatial multiplexing.

2. Calculate effective throughputs for all four configurations.

Given:

- Baseline spectral efficiency = 2 bits/s/Hz
- Proposed spectral efficiency = 4 bits/s/Hz
- Overhead = 20%

For this comparison, the assumed simultaneous layers are 2 for configuration A, 4 for configurations B and C using MU-MIMO, and 8 ideal layers for configuration D.
Configuration A: 20 MHz, 2×2 MIMO
$$ T=20\times2\times2\times0.8 $$
$$ \boxed{T=64\ Mbps} $$
Configuration B: 40 MHz, 4×4 MIMO
$$ T=40\times4\times4\times0.8 $$
$$ \boxed{T=512\ Mbps} $$
Configuration C: 80 MHz, 4×4 MIMO
$$ T=80\times4\times4\times0.8 $$

$$\boxed{T=1024\ Mbps}$$
Configuration D: 100 MHz, 8×8 MIMO
$$T=100\times4\times8\times0.8$$
$$\boxed{T=2560\ Mbps}$$
Comparison table

Configuration	Bandwidth	MIMO	Effective throughput
A	20 MHz	2×2	64 Mbps
B	40 MHz	4×4	512 Mbps
C	80 MHz	4×4	1024 Mbps
D	100 MHz	8×8	2560 Mbps

These are theoretical results under the stated spatial-layer assumptions, not measured network results.

3. Evaluate whether each configuration meets the aggregate demand of 1 Gbps.

Required throughput:
$$T_{\text{demand}}=1000\ Mbps$$

Configuration	Throughput	Meets demand?
A	64 Mbps	No
B	512 Mbps	No
C	1024 Mbps	Yes
D	2560 Mbps	Yes

Result: Configurations C and D satisfy the 1 Gbps demand under ideal assumptions.

4. Select a configuration and explain the trade-off.

Selected configuration: Configuration C (80 MHz, 4×4 MIMO).

Reasons:

1. It provides 1024 Mbps throughput.
2. It satisfies the 1 Gbps user demand.
3. It uses less bandwidth than configuration D.
4. It supports MU-MIMO for serving multiple users simultaneously.
5. It has lower hardware complexity than 8×8 MIMO.

Trade-offs:

- Higher bandwidth improves throughput but consumes more spectrum.
- Higher-order MIMO improves capacity but increases hardware complexity.
- MU-MIMO improves the number of simultaneously served users.
- Configuration D provides higher theoretical throughput but requires more antennas and processing resources.

Conclusion: Configuration C provides a good balance between throughput, spectrum utilization and implementation complexity.

Note: The supplied three-component-carrier example gives 60 MHz nominal bandwidth and 54 MHz usable bandwidth at 90% efficiency. It is a separate CA case from the 80 MHz candidate configuration.

CO-2 Assignment Question Paper eca5604.pdf

Stage 3 – Beamforming Design and Link-Budget Validation (20 Marks)

1. Calculate wavelength and half-wavelength antenna spacing at 3.5 GHz.

Given:

- Frequency = 3.5 GHz
- Speed of light = 3×10^8 m/s

Formula:
$$\lambda=\frac{c}{f}$$
Substitute:
$$\lambda=\frac{3\times10^8}{3.5\times10^9}$$
$$\boxed{\lambda=0.0857\ m}$$
Half-wavelength spacing:
$$d=\frac{\lambda}{2}$$
$$d=\frac{0.0857}{2}$$
$$\boxed{d=0.04285\ m=4.285\ cm}$$
Result: The wavelength is 8.57 cm and the antenna spacing is approximately 4.285 cm.

2. Calculate ideal array gain and predicted received power.

Given:

- Single-element received power = −100 dBm
- Formula:

$$G=10\log_{10}(N)$$
Received power:
$$P_r=P_{\text{single}}+G$$
For 4 antenna elements:
$$G=10\log_{10}(4)=6.02\ dB$$
$$P_r=-100+6.02$$
$$\boxed{P_r=-93.98\ dBm}$$
For 16 antenna elements:
$$G=10\log_{10}(16)=12.04\ dB$$
$$P_r=-100+12.04$$
$$\boxed{P_r=-87.96\ dBm}$$
For 64 antenna elements:
$$G=10\log_{10}(64)=18.06\ dB$$
$$P_r=-100+18.06$$
$$\boxed{P_r=-81.94\ dBm}$$
Comparison table

Antenna elements	Array gain	Received power
4	6.02 dB	−93.98 dBm
16	12.04 dB	−87.96 dBm
64	18.06 dB	−81.94 dBm

3. Evaluate which arrays satisfy the target received-power requirement.

Target received power:
$$P_r\geq-90\ dBm$$

- 4 elements: −93.98 dBm – Does not satisfy.
- 16 elements: −87.96 dBm – Satisfies.
- 64 elements: −81.94 dBm – Satisfies.

Result: The 16-element and 64-element antenna arrays satisfy the target requirement under ideal conditions.

4. Design a simulation to compare beam patterns and beam steering.

Beamforming directs the transmitted signal towards a particular user to improve signal strength.

Simulation procedure:

1. Create a uniform linear antenna array.
2. Use 4, 16 and 64 antenna elements.
3. Set the antenna spacing to half wavelength.
4. Calculate the beam pattern.
5. Change the steering angle.
6. Introduce phase errors to study practical performance.
7. Compare the beam patterns.

Non-ideal factor: Phase errors can reduce the beamforming gain and affect signal direction.

5. Justify the selected array.

Selected array: 16-element antenna array.

Reasons:

1. It provides received power of −87.96 dBm.
2. It satisfies the −90 dBm target.
3. It provides better beamforming gain than the 4-element array.
4. It has lower hardware complexity than the 64-element array.
5. It provides a reasonable balance between performance and cost.

Conclusion: The 16-element array is suitable because it satisfies the received-power requirement with moderate implementation complexity.

Stage 5 – Simulation Validation, Performance Comparison and Engineering Decision (20 Marks)

1. Implement baseline and proposed configurations in Python.

Software used: Python

Libraries required:

- NumPy – Mathematical calculations.
- Matplotlib – Performance graphs.

Installation:

pip install numpy matplotlib

Simulation input parameters:

Parameter	Baseline	Proposed
Frequency	3.5 GHz	3.5 GHz
Bandwidth	20 MHz	80 MHz
MIMO	2×2	4×4
Spatial layers	1	4
Spectral efficiency	2	4
Overhead	20%	20%
Users	100	100

Simulation procedure:

1. Define the baseline and proposed network parameters.
2. Calculate effective throughput.
3. Calculate spectral efficiency.
4. Calculate antenna array gain.
5. Evaluate received power.
6. Change bandwidth, MIMO streams and channel conditions.
7. Generate four performance graphs.

8. Compare both network configurations.

2. Generate and interpret four performance plots.

The following four plots are generated using the Python program provided below.

Plot 1: Effective Throughput versus Bandwidth

This graph shows how throughput increases when bandwidth increases.

Interpretation: Higher bandwidth provides higher data transmission capacity.

Plot 2: Throughput versus Number of MIMO Streams

This graph compares throughput using different numbers of spatial streams.

Interpretation: Increasing the number of spatial streams improves network throughput under suitable channel conditions.

Plot 3: Received Power versus Number of Antenna Elements

This graph compares received power for 4, 16 and 64 antenna elements.

Interpretation: Increasing antenna elements improves beamforming gain and received signal power under ideal conditions.

Plot 4: Performance under Different Channel and Interference Conditions

This graph compares throughput under good, moderate and poor channel conditions.

Interpretation: Poor channel quality and interference reduce network throughput.

3. Compare baseline and proposed designs using four metrics.

Performance metric	Baseline	Proposed
Bandwidth	20 MHz	80 MHz
MIMO	2×2	4×4
Effective throughput	32 Mbps	1024 Mbps
Effective spectral efficiency	1.6 bits/s/Hz	12.8 bits/s/Hz
Received power	−100 dBm*	−87.96 dBm*
User demand	1000 Mbps	1000 Mbps
Demand satisfied	No	Yes

*Received-power figures use the single-element baseline and ideal 16-element proposed-array model. The spectral-efficiency values include the assumed spatial layers and 20% overhead.

Result: The proposed network provides significantly higher theoretical throughput than the baseline network and satisfies the aggregate user demand under ideal assumptions.

4. Conduct sensitivity analysis.

Sensitivity analysis checks how changes in network parameters affect performance.

A. Change in overhead

Overhead	Proposed throughput
10%	1152 Mbps
20%	1024 Mbps
25%	960 Mbps
30%	896 Mbps

Observation: When overhead increases to 25%, throughput falls below 1 Gbps.

B. Change in channel quality

Channel quality factor	Throughput
1.0	1024 Mbps
0.9	921.6 Mbps

0.8	819.2 Mbps
0.7	716.8 Mbps

Observation: Poor channel conditions reduce throughput. At a channel quality factor of 0.9, the proposed network no longer satisfies the required 1 Gbps.

Conclusion: The proposed network performs well under ideal conditions, but additional optimization is required under poor channel conditions and high overhead.

5. Recommend the final network configuration.

Recommended configuration:

- Operating frequency: 3.5 GHz
- Bandwidth: 80 MHz
- MIMO: 4×4
- Spatial multiplexing: 4 MU-MIMO user layers
- Beamforming: 16-element antenna array
- Overhead: 20%

Justification:

1. Provides theoretical throughput of 1024 Mbps.
2. Meets the aggregate user demand of 1 Gbps under ideal assumptions.
3. Improves received signal power through beamforming.
4. Supports multiple users using MU-MIMO.
5. Uses less bandwidth and has lower antenna complexity than the 100 MHz, 8×8 alternative.

Limitations:

- Actual throughput may be lower because of interference.
- Hardware implementation increases cost.
- Channel quality affects MIMO performance.
- Beamforming requires accurate phase control.

Future improvements:

- Use advanced interference management.
- Improve channel estimation.
- Use adaptive beamforming.
- Optimize resource allocation.
- Introduce advanced MIMO techniques.

Final conclusion: The proposed 5G NR network using 80 MHz bandwidth, 4×4 MIMO and a 16-element beamforming array provides a suitable balance between throughput, signal quality, spectrum utilization and implementation complexity.

Python Simulation Program

Save this program as 5g_simulation.py.

import numpy as npplt.savefig("plot3_antenna.png")plt.show()# Plot 4: Throughput under different channel conditionschannel = np.array([1.0, 0.9, 0.8, 0.7])base_results = [throughput(20, 2, 2, c) for c in channel]proposed_results = [throughput(80, 4, 4, c) for c in channel]plt.figure()plt.plot(channel, base_results, marker='o', label="Baseline Network")plt.plot(channel, proposed_results, marker='s', label="Proposed Network")plt.xlabel("Channel Quality Factor")plt.ylabel("Throughput (Mbps)")plt.title("Performance under Different Channel Conditions")plt.legend()plt.grid()plt.savefig("plot4_channel.png")plt.show()# Sensitivity analysis: overheadprint("\nOVERHEAD SENSITIVITY")for o in [0.10, 0.20, 0.25, 0.30]: result = prop_bw *

prop_se * prop_layers * (1-o) print(o * 100, "% overhead:", result, "Mbps")# Final comparisonprint("\nFINAL COMPARISON")print("Baseline:", baseline, "Mbps")print("Proposed:", proposed, "Mbps")print("Required:", demand, "Mbps")if proposed >= demand: print("Proposed network meets user demand.")else: print("Proposed network does not meet user demand.")

Expected output

BASELINE THROUGHPUT: 64.0 Mbps
PROPOSED THROUGHPUT: 1024.0 Mbps

OVERHEAD SENSITIVITY
10.0 % overhead: 1152.0 Mbps
20.0 % overhead: 1024.0 Mbps
25.0 % overhead: 960.0 Mbps
30.0 % overhead: 896.0 Mbps

FINAL COMPARISON
Baseline: 64.0 Mbps
Proposed: 1024.0 Mbps
Required: 1000 Mbps
Proposed network meets user demand.
The graphs are generated and saved as four PNG files in the same folder as the Python program.

1. 5G NR Architecture and RAN Functional Split

5G NR architecture consists of User Equipment (UE), Next Generation Radio Access Network (NG-RAN) and 5G Core Network (5GC).
Main components:

1. UE: Mobile phones and other devices used to access the network.
2. gNodeB: Provides wireless communication between users and the core network.
3. AMF: Handles user registration and mobility management.
4. SMF: Manages PDU sessions.
5. UPF: Transfers user data between the network and external networks.
6. AUSF: Performs user authentication.
7. UDM: Stores subscriber information.
8. PCF: Provides network policy control.

RAN functional split:

- Central Unit (CU): Handles higher-level network functions.
- Distributed Unit (DU): Handles real-time radio processing.
- Radio Unit (RU): Handles radio transmission and reception.

Conclusion: 5G NR architecture improves network flexibility, scalability and communication performance.

2. RRC Operation and Connection Establishment

RRC stands for Radio Resource Control. It manages communication between the mobile device and the base station.
RRC states:

1. RRC_IDLE: The device is not actively connected to the network.

2. RRC_INACTIVE: The device maintains some network context while reducing power consumption.
3. RRC_CONNECTED: The device is actively connected and can exchange data.

Connection establishment steps:

1. The mobile device searches for a suitable 5G cell.
2. It performs random access.
3. It sends an RRC connection request.
4. The base station sends an RRC setup message.
5. The device sends an RRC setup complete message.
6. The network performs authentication and security procedures.
7. The device becomes ready for data communication.

Conclusion: RRC manages connection establishment, mobility and radio resource control in 5G networks.